

Low vs. higher-dose dark chocolate and blood pressure in cardiovascular high-risk patients.

BACKGROUND: Dark chocolate may have blood pressure-lowering properties. We conducted a prospective randomized open-label blinded end-point design trial to study a potential dose dependency of the presumed antihypertensive effect of dark chocolate by directly comparing low vs. higher doses of dark chocolate over the course of 3 months. **METHODS:** We enrolled a total of 102 patients with prehypertension/stage 1 hypertension and established cardiovascular end-organ damage or diabetes mellitus. Patients were randomly assigned to receive either 6 or 25 g/day of flavanol-rich dark chocolate for 3 months. The difference in 24-h mean blood pressure between groups was defined as the primary outcome measure. **RESULTS:** Significant reductions in mean ambulatory 24-h blood pressure were observed between baseline and follow-up in both groups (6 g/day: -2.3 mm Hg, 95% confidence interval -4.1 to -0.4; 25 g/day: -1.9 mm Hg, 95% confidence interval -3.6 to -0.2). There were no significant differences in blood pressure changes between groups. In the higher-dose group, a slight increase in body weight was noted (0.8 kg, 95% confidence interval 0.06 to 1.6). **CONCLUSIONS:** The findings are consistent with the hypothesis that dark chocolate may be associated with a reduction in blood pressure (BP). However, due to the lack of a control group, confounding may be possible and the results should be interpreted with caution.